

Electric Vehicle Load Forecasting Based on Probabilistic Modeling and Deep Learning

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Abstract—With the large-scale integration of electric vehicles (EVs) and the rising penetration levels, the stochasticity, intermittency, and spatiotemporal coupling of charging loads bring considerable challenges to grid planning and operation. To address these issues, this paper proposes a probabilistic modeling and deep learning-based approach for EV charging load forecasting. A time-varying origin-destination (OD) probability matrix is first constructed using dynamic traffic network topology and real-time flow-speed relationships, with Dijkstra's algorithm employed to simulate optimal routes. Monte Carlo sampling is then applied to departure times, driving parameters, dwelling durations, and charging decisions to generate historical load data that capture spatiotemporal patterns. On this basis, a hybrid CNN-BiLSTM model is developed, where convolutional networks extract spatial features and bidirectional long short-term memory networks capture temporal dependencies, enabling accurate probabilistic forecasting of complex charging loads.

Keywords—electric vehicles, probabilistic modeling, deep learning, shortest path planning, charging load forecasting

I. INTRODUCTION

In recent years, the rapid growth of the electric vehicle (EV) industry has drawn increasing attention to its impact on power grids. Large-scale EV integration supports clean energy utilization and energy transition, but also introduces new load characteristics. Unlike traditional loads, EV charging is random, intermittent, and clustered, depending strongly on users' travel patterns and daily schedules, which often leads to concentrated demand in specific periods and regions. This can cause sharp load fluctuations in distribution networks and, without accurate forecasting, may induce power flow redistribution, voltage fluctuations, harmonic distortion, local overloads, greater peak-to-valley differences, and reduced power quality. Such uncertainties place higher demands on network planning, operation, and demand response. Therefore, high-precision forecasting of EV charging loads is essential for ensuring secure

grid operation and promoting the coordinated development of transportation electrification and low-carbon energy [1–4].

EV load forecasting is still evolving, with ongoing efforts to improve prediction accuracy and real-time scheduling. Researchers have proposed diverse modeling methods, algorithms, and data processing techniques, achieving significant progress. Reference [5] analyzed the National Household Travel Survey (NHTS) dataset to identify temporal patterns of EV charging loads, used autoencoders to extract travel features, calculated EV arrival probabilities, and established a spatiotemporal charging load model. Reference [6] examined state of charge (SOC) characteristics, travel time distributions, and price sensitivity using historical charging data and V2G surveys, applied a random forest for feature importance analysis, and employed Monte Carlo simulations to generate travel and charging and discharging scenarios. Reference [7] utilized Latin Hypercube Sampling for initial departure times and SOC, developed a fuzzy evaluation-based charging behavior model, incorporated traffic conditions, and dynamically simulated travel paths to derive spatiotemporal charging demand. Reference [8] combined fuzzy clustering with a BP neural network to extract key factors for different load days and predict daily charging demand. Reference [9] applied an improved gravity model based on urban functional zoning to construct travel matrices, sampled EV travel mileage with SOC, and developed a spatiotemporal forecasting model. Reference [10] proposed a real-time dynamic network model integrating traffic data, considering link and node impedances, and used a dynamic Dijkstra algorithm with origin-destination (OD) matrices to simulate random EV movements and plan charging paths. Reference [11] introduced a Convolutional Neural Network (CNN)-based forecasting method optimized with the Lion Algorithm and a Niche immunity mechanism, enhancing prediction accuracy and stability for charging station loads.

Overall, existing studies mainly rely on historical datasets, travel probability models, dynamic traffic simulations, or user

behavior modeling, often using single machine learning techniques. However, they face limitations in capturing complex travel patterns, characterizing spatiotemporal load features, and achieving high prediction accuracy. Moreover, due to the relatively short history of EVs, real charging data are limited and often proprietary, leading some studies to depend on synthetic or survey-based datasets, which reduces generalization and practical applicability. Further improvements are needed in modeling nonlinear load characteristics, exploring multidimensional coupling relationships, and enabling high-precision dynamic forecasting.

To address these challenges, this paper proposes an EV charging load forecasting method that integrates probabilistic modeling with deep learning. By fully considering the spatiotemporal distribution of charging loads, a probability-based EV load model is constructed using dynamic traffic flow data and travel probability analysis. This model generates substantial historical load data to train neural networks, significantly enhancing forecasting accuracy and efficiency, and providing reliable support for grid dispatch and charging infrastructure planning.

II. TRAFFIC NETWORK MODEL

A. Traffic Network Topology Structure

Based on graph theory principles, the traffic network can be abstracted into a simplified topological structure, namely the traffic network topology graph $G(V, E)$, where V represents the set of network nodes, and E represents the set of edges between nodes. Taking the topology graph shown in Fig. 1 as an example, its corresponding topological structure can be represented by the adjacency matrix D , as shown in Table I. The elements in matrix D reflect the road lengths between different nodes, thus intuitively depicting the topological features of the traffic network.

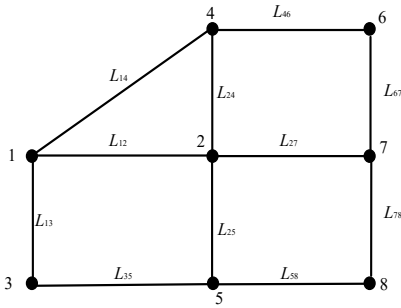


Fig. 1. Transportation network topology.

TABLE I. ROAD NETWORK TOPOLOGY MATRIX

Node	1	2	3	4	5	6	7	8
1	0	L_{12}	L_{13}	L_{14}	inf	inf	inf	inf
2	L_{12}	0	inf	L_{24}	L_{25}	inf	L_{27}	inf
3	L_{13}	inf	0	inf	L_{35}	inf	inf	inf
4	L_{14}	L_{24}	inf	0	inf	L_{46}	inf	inf
5	inf	L_{25}	L_{35}	inf	0	inf	inf	L_{58}
6	inf	inf	inf	L_{46}	inf	0	L_{67}	inf
7	inf	L_{27}	inf	inf	inf	L_{67}	0	L_{78}
8	inf	inf	inf	inf	L_{58}	inf	L_{78}	0

$$D_{ij} = D_{ji} = \begin{cases} L_{ij} & (i, j) \text{ is an edge in } E(G) \\ 0 & i = j \\ \text{inf} & (i, j) \text{ is not an edge in } E(G) \end{cases} \quad (1)$$

where, L_{ij} represents the length of the road section between node i and node j ; inf indicates no direct connection. Based on the network adjacency matrix D , path simulation algorithms can be efficiently used to find the shortest path between any two nodes.

B. Dynamic Speed-Flow Relationship Model

The driving speed of EVs directly affects energy consumption per unit distance and is a crucial factor determining charging demand levels. Therefore, reasonably characterizing the coupling relationship between vehicle speed and traffic flow is key to EV charging load forecasting. Considering that vehicle speed in urban traffic environments is significantly constrained by road capacity and traffic flow saturation, this paper constructs a dynamic speed-flow relationship model based on traditional traffic flow theory to reflect the variation patterns of vehicle driving characteristics under different traffic conditions [12].

At time t , the average speed $v_{ij,t}$ of an EV on road L_{ij} is expressed as:

$$v_{ij,t} = \frac{v_{ijf}}{1 + \left(\frac{Q_{ij,t}}{C_{ij}}\right)^\beta} \quad (2)$$

where, v_{ijf} is the free-flow speed of road L_{ij} ; C_{ij} is the maximum capacity of the road; $Q_{ij,t}$ is the traffic flow on road L_{ij} at time t ; $Q_{ij,t} / C_{ij}$ represents the flow-to-capacity ratio, measuring the saturation degree of the road. The parameter β is an adjustment factor, given by:

$$\beta = a + b\left(\frac{Q_{ij,t}}{C_{ij}}\right)^n \quad (3)$$

where, a , b , and n are adaptive coefficients set according to road grade and traffic conditions. For different road grades (e.g., highways, arterials, branches), their values can be calibrated using measured data.

This model effectively captures the nonlinear pattern of speed gradually decreasing with increasing traffic flow, enabling a dynamic description of EV travel characteristics.

III. CHARGING LOAD PROBABILITY FORECASTING MODEL

A. OD Travel Probability Matrix

The travel probability matrix is a stochastic transition matrix constructed based on OD flow data, used to describe the probability distribution characteristics of vehicle movement between nodes in a transportation network [13]. The row elements of this matrix reflect the probability weights of vehicles diverting from a specific origin to various destination nodes, while the column elements represent the attraction probability of each destination node to flows from different origins. By obtaining real-time traffic flow observation data

from key detection sections of the network, probabilistic transition relationships for multiple types of EVs in spatial and temporal dimensions can be constructed.

The core of building the travel probability matrix lies in establishing the mapping relationship between observed flow values and OD demand. Setting the objective function to minimize the deviation between observed and estimated flows, the following optimization model is established:

$$\min F = \sum_{k=1}^L \sum_{h=1}^H \left(q_{k,h} - \sum_{i=1}^N \sum_{j=1}^N \gamma_{ij,k} \cdot T_{ij,h} \right)^2 \quad (4)$$

$$s.t. \quad T_{ij,h} \geq 0 \quad \forall i, j, h$$

where, $q_{k,h}$ is the observed traffic flow (veh/h) at detection section k in time period h ; N is the number of traffic zones; L is the total number of detection sections; $T_{ij,h}$ is the vehicle travel demand from zone i to zone j in period h ; $\gamma_{ij,k}$ is the path assignment coefficient, representing the proportion of traffic flow from zone i to zone j that passes through detection section k . This coefficient is determined by traffic assignment theory; it is 1 if the optimal path for OD pair (i, j) passes through section k , otherwise 0. This study uses a multi-path probability assignment method, where $\gamma_{ij,k}$ is a continuous value between 0 and 1, negatively correlated with path impedance. Based on the user equilibrium assignment principle, the iterative weighted least squares method is used to solve the above optimization problem, obtaining the time-varying OD demand matrix $T_{ij,h}$. To improve solution accuracy, prior OD information is introduced as a constraint.

Using the back-calculated full-period OD demand matrix for multiple types of EVs, the corresponding travel probability matrix can be obtained through row normalization. For the travel probability matrix of vehicle type m in time period h , the calculation formula is:

$$P_{ij,m,h} = \frac{T_{ij,m,h}}{\sum_{i=1}^N \sum_{j=1}^N T_{ij,m,h} + \varepsilon} \quad 1 \leq i, j \leq N \quad (5)$$

where, $T_{ij,m,h}$ represents the travel probability of vehicle type m from origin i to destination j in period h ; ε is a smoothing parameter used to avoid division by zero numerical issues. When $i = j$, $P_{ij,m,h}$ represents the probability of a vehicle staying at node i , reflecting the traffic holding capacity of that node. Using equation (5), the travel probability matrices for different types of EVs for each hour of a 24-hour day can be obtained, providing basic data support for subsequent vehicle behavior prediction and charging demand analysis.

B. Driving Path Simulation and Planning

In the theory of optimal path search based on graph-structured transportation networks, the Dijkstra algorithm, as a classic greedy strategy algorithm [14], can effectively identify the minimum cost path from a source node to a target node. Let G represent the set of edge sequences contained in a candidate path, as shown in equation (6), where L represents the total

length of path G . The optimization goal of this algorithm is to identify the optimal solution $\bar{L}$ that achieves the global minimum value of L from the entire feasible path space.

$$L = \sum_{(i,j) \in G} L_{ij} \quad (6)$$

$$\begin{cases} T_{ij,t} = \sum_{k \in \{1,2,3\}} \frac{L_{ij,k}}{V_{ij,k,t}} \\ L'_t = \sum_{(i,j) \in G} T_{ij,t} \end{cases} \quad (7)$$

Based on the spatiotemporal path simulation model constructed in equation (7), the dynamic estimation of the complete travel time for path segment (i, j) can be achieved. Within this framework, the link travel time is used instead of the traditional geometric distance as the weight factor in the adjacency matrix, and minimizing the total time cost expressed in the equation is set as the optimization criterion for the Dijkstra algorithm, thus constructing a time-optimal path selection strategy. When a vehicle arrives at an intersection node, the system re-optimizes the subsequent journey based on real-time traffic state information to ensure the timeliness of path selection.

For vehicles in dynamic motion, estimating the remaining journey energy consumption E' requires considering the vehicle's current location information. Suppose at time t , the vehicle is at a certain position on the road section (p, q) , its spatial distance from the starting node p is denoted as $D_{pq,t}$, the current road section is marked as $K = k$, and the time-optimal path from node p to the destination q is G' . The expected energy consumption E'_t for the vehicle's subsequent journey can be quantified by the following equation:

$$E'_t = \left(\sum_{k=1}^K L_{pq,k} - D_{pq,t} \right) \omega_{pq,k,t} + \sum_{k=K+1}^3 L_{pq,k} \omega_{pq,k,t} + \sum_{\substack{k \in \{1,2,3\} \\ (i,j) \in G', (i,j) \neq (p,q)}} L_{ij,k} \omega_{pq,k,t} \quad (8)$$

where, $\omega_{pq,k,t}$ represents the EV driving mileage energy consumption, the first term on the right quantifies the expected energy contribution of the remaining part of the current road section, the second term calculates the energy demand of subsequent road sections in the planned path sequence, and the third term covers the total energy consumption level of other roads in the optimal path excluding the current road section.

C. Spatiotemporal Sampling of Charging Demand

Based on the established traffic network model and charging load probability model, Fig.2 shows the flowchart for the spatiotemporal distribution sampling analysis of EV charging demand within a region [12].

The steps for establishing the probabilistic forecasting load curve for EVs based on this analysis flowchart are as follows:

1) Determine the spatial distribution quantity of various types of EVs at each traffic network node, establishing a basic

relevant dataset for EVs covering each traffic node in the study area.

2) Use the Monte Carlo random sampling method to generate driving characteristic parameters and charging characteristic parameters for each EV.

3) Based on the generated two types of key characteristic parameters, simulate driving paths using the dynamic Dijkstra algorithm to search for the path with the shortest travel time within the OD travel space for each EV. Vehicles travel according to the planned path and update driving and charging parameters in real-time.

4) Determine charging demand based on the overall driving path and parameters, calculate the corresponding charging duration for each EV, and accumulate and superimpose these to obtain the regional EV charging load curve for different time periods.

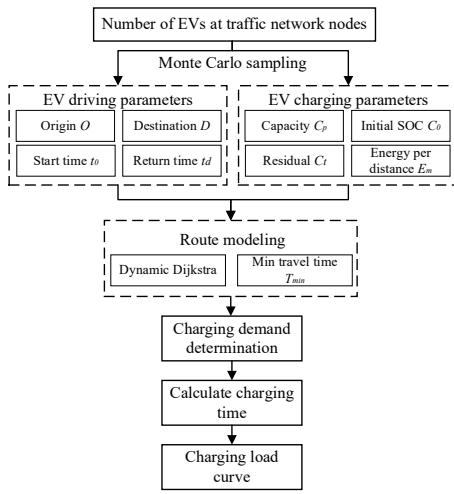


Fig.2. Flow chart of the EV charging demand prediction.

IV. CNN-BiLSTM-BASED EV LOAD FORECASTING MODEL

EV charging demand is characterized by strong randomness, non-stationary distribution patterns, and distinct periodicity. Since charging behavior is influenced by multiple factors (such as user travel habits, the spatial distribution of charging facilities, and electricity pricing policies), the resulting load distribution exhibits complex spatiotemporal coupling characteristics. Moreover, the contribution of these influencing factors to prediction accuracy varies considerably. With the rapid expansion of the EV fleet, traditional probability sampling-based forecasting models face growing challenges, as computational complexity rises sharply in large-scale forecasting tasks, making it difficult to satisfy real-time requirements. Consequently, there is an urgent need to adopt more efficient and intelligent algorithms.

Since the training of deep learning models requires sufficient historical load samples, and measured EV charging load data are relatively scarce, this study adopts simulation data generated by a probability sampling-based EV load forecasting model as the primary data source. On this basis, a deep neural network forecasting framework integrating a CNN-BiLSTM mechanism is developed. The model captures the intrinsic relationship

between user travel patterns and charging decisions, establishes a nonlinear mapping from behavioral features to load demand, and provides a deep learning-based solution for EV charging load forecasting. The detailed network architecture is illustrated in Fig.3 [15].

A. CNN-BiLSTM Load Forecasting Model

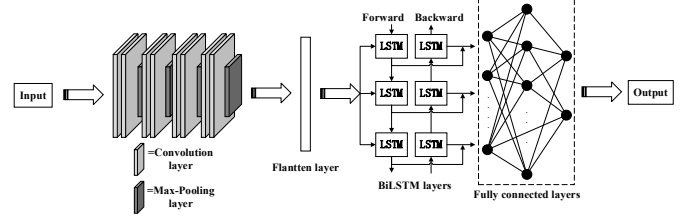


Fig.3. CNN-BiLSTM Model structure diagram.

1) The second layer is the CNN layer. This layer effectively captures local spatial dependencies in the input feature sequence and, when combined with the subsequent Bidirectional Long Short-Term Memory network (BiLSTM), compensates for the latter's limitations in extracting spatial features while preserving the continuity and correlation of the time series. After entering the CNN layer, the input data undergoes convolution, pooling, and feature dimensionality reduction operations. For time series data, one-dimensional convolution is typically applied, where a fixed-length kernel slides along the time axis to extract features. If the kernel size is set to λ the convolution operation is performed on the segment from time step u to $(u + \lambda - 1)$ with a stride of 1. The weight matrix is denoted as W , with dimensions $n \times \lambda$, and the bias term is denoted as b . In this process, feature mapping is applied to the sequence segment within the time window, and the i -th feature output o_u is calculated as:

$$o_u = f(W \otimes x_{u:u+\lambda-1} + b) \quad (9)$$

where, $f(\cdot)$ is the nonlinear activation function and denotes the convolution operation, $\otimes$ represents the convolution operation. As the kernel slides over the sequence, the outputs form a feature matrix of size $(t - \lambda + 1) \times 1$ where t is the sequence length, λ is the kernel size, and 1 is the number of kernels.

$$O = [o_1, o_2, \dots, o_{t-\lambda+1}] \quad (10)$$

The CNN layer typically contains multiple convolution kernels, allowing the extraction of multiple sets of local feature representations. The features are then compressed and reduced in dimensionality through a pooling operation. For example, using max pooling with a window size of 2 and a stride of 2, a feature matrix of size $[(t - \lambda + 1) / 2] \times 1$ can be obtained.

$$o_u = \max(o_{2u-1}, o_{2u}), \quad u = 1, 2, \dots \quad (11)$$

This process not only retains the main feature information but also effectively reduces redundancy and improves computational efficiency. Finally, the spatial features extracted by the CNN layer are integrated into the BiLSTM network for continued modeling and prediction of temporal dependencies.

2) The third layer is the BiLSTM layer. This layer consists of two LSTM sub-networks, forward and reverse, which can simultaneously capture forward and backward temporal dependencies in the time dimension, enabling bidirectional

modeling of the charging load sequence. The forward sub-network processes the input sequence in chronological order, while the reverse sub-network processes it in reverse chronological order.

In the BiLSTM structure, each LSTM unit contains gating mechanisms (input gate, forget gate, and output gate) and state storage mechanisms (candidate state, cell state, and hidden state). The cell state maintains long-term memory information, while the hidden state reflects short-term dynamic features. For the input at time t , BiLSTM integrates the forward hidden state S_t^{forward} and the backward hidden state S_t^{backward} through a concatenation operation to obtain the final bidirectional hidden state S_t , calculated as:

$$S_t = [S_t^{\text{forward}} \oplus S_t^{\text{backward}}] \quad (12)$$

where, $\oplus$ denotes the vector concatenation operation. Through this bidirectional structure, BiLSTM can simultaneously utilize both historical and future information in the time dimension, enabling more comprehensive modeling of the complex dynamic characteristics of the load sequence.

B. EV Load Forecasting

The EV charging load data based on the aforementioned probability forecasting is used as training data input into the CNN-BiLSTM model for learning and prediction. The specific prediction process is shown in Fig.4.

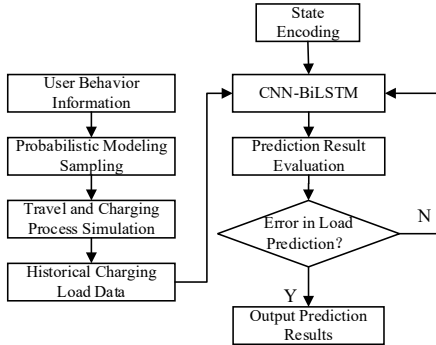


Fig.4. CNN-BiLSTM Load forecasting process diagram.

After state encoding and user behavior information modeling, the sample data enters the load forecasting process. First, based on user travel and charging behavior, probability modeling methods are used for sampling to generate simulated data consistent with user travel habits. This is combined with historical charging load data to form complete input samples. Subsequently, the input data enters the CNN-BiLSTM prediction model. The CNN layer effectively extracts local spatial features from the input sequence and achieves dimensionality reduction through convolution and pooling operations. The BiLSTM layer, utilizing its bidirectional memory structure (forward and reverse), fully captures the bidirectional temporal dependencies of the load data, thereby enhancing feature representation capability. After deep feature extraction and sequence modeling, the model outputs the prediction results.

In the prediction stage, the output results undergo an error evaluation process and are compared with a preset error

threshold. If the prediction error meets the accuracy requirement, the model outputs the final prediction result; if the error exceeds the threshold, it is necessary to return to the model for further parameter updates and optimization to ensure the reliability and stability of prediction performance.

Through the above process, the CNN-BiLSTM model, based on user behavior modeling and supported by historical load data, achieves accurate prediction of future charging loads, providing effective support for the rational scheduling of EV charging demand and grid operation optimization.

V. CASE STUDY ANALYSIS

A. Case System Overview

This paper introduces probabilistic modeling methods to simulate vehicle travel and charging behaviors. Specifically, Monte Carlo sampling is employed to generate travel and charging scenarios for a fleet of electric vehicles, with a fleet size of 10,000 vehicles. Charging processes are simulated over a 30-day time domain to obtain daily charging load data characterized by randomness and variability. Subsequently, the data is aggregated at 15-minute intervals to construct a charging load sequence $\{X_1, X_2, \dots, X_T\}$ with dimensions 96×100 , providing input data for training deep learning models.

Before model training, all load sequences undergo Z-score standardization processing:

$$\bar{X} = \frac{1}{T} \sum_{t=1}^T X_t \quad (13)$$

$$X'_t = \frac{X_t - \bar{X}}{\sigma} \quad (14)$$

$$\sigma = \sqrt{\frac{1}{T} \sum_{t=1}^T (X_t - \bar{X})^2} \quad (15)$$

Through the above normalization operation, standardized sequences with a mean of 0 and a variance of 1 can be obtained, avoiding training bias caused by differences in data scales.

In the forecasting task, the training set consists of 24-hour load data from the previous 30 days, while the test set corresponds to the load data of the 31st day, forming the basis of the prediction experiment. During forecasting, a sliding time window approach is employed to generate input-output pairs, where a historical load sequence of length 96 serves as the input variable $X \in R^{T \times I}$, and the load at the next time step serves as the target $Y \in R^{I \times I}$, enabling short-term rolling prediction.

To evaluate the effectiveness of the proposed CNN-BiLSTM model, two comparative models are set: a single LSTM network and a CNN-LSTM hybrid model. All algorithms are implemented on the MATLAB platform and trained and tested on the same dataset under identical parameter settings. The performance comparison on the test set allows for a comprehensive assessment of the prediction accuracy and generalization ability of the model after incorporating the attention mechanism.

B. Case Result Analysis

The EV load forecasting results are shown in Fig.5, which presents a comparison of the prediction curves of the three

neural network models under the same dataset. It can be intuitively seen from the figure that different models vary in their ability to capture the changing trends of the EV load and fit the actual fluctuation characteristics. To further quantitatively evaluate the forecasting performance of each model, this paper selects multiple evaluation metrics for comparative analysis, and the results are shown in Table II. These metrics can more comprehensively reflect the advantages and disadvantages of each model in terms of prediction accuracy, stability, and response capability to load fluctuations, thus providing a reference for the optimization and improvement of EV load forecasting methods.

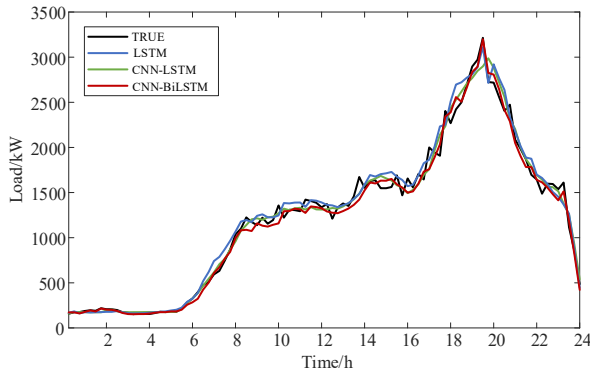


Fig.5. Comparison chart of load forecasting results.

TABLE II. MODEL PERFORMANCE EVALUATION INDICATORS

Model	RMSE(kW)	R ²	MAPE(%)
LSTM	107.07	0.9824	6.726
CNN-LSTM	95.10	0.9861	5.326
CNN-BiLSTM	78.20	0.9906	4.869

From the results in Table 2, it can be seen that all three models can satisfactorily complete the EV charging load forecasting task, with the CNN-BiLSTM model performing the best. Compared to LSTM, CNN-LSTM shows improvements in RMSE, MAPE, and R², indicating that the convolutional structure can effectively enhance feature extraction capability. Furthermore, CNN-BiLSTM significantly reduces the prediction error on this basis. Compared to CNN-LSTM, RMSE decreases by approximately 16.90 kW, MAPE decreases by about 0.457 percentage points, and R² increases by 0.0045, verifying its advantage in capturing bidirectional temporal dependencies, thereby significantly improving prediction accuracy.

VI. CONCLUSION

This paper proposes an EV load forecasting method that integrates probabilistic modeling and deep learning. By introducing a traffic network model and travel probability modeling feature analysis, it effectively characterizes the randomness and uncertainty of EV charging loads. This method uses probabilistic modeling to enhance the ability to express load fluctuation patterns and combines the feature extraction and nonlinear fitting advantages of deep learning neural networks to achieve high-precision prediction of complex time series data. Simulation results show that compared to single deep learning models, the method integrating probabilistic modeling performs

better in both prediction accuracy and robustness. Its evaluation metrics are superior to those of the comparison models, maintaining strong generalization capability while handling random load fluctuations. Therefore, this method provides an effective and feasible new approach for EV charging load forecasting.

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